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Correct Answer

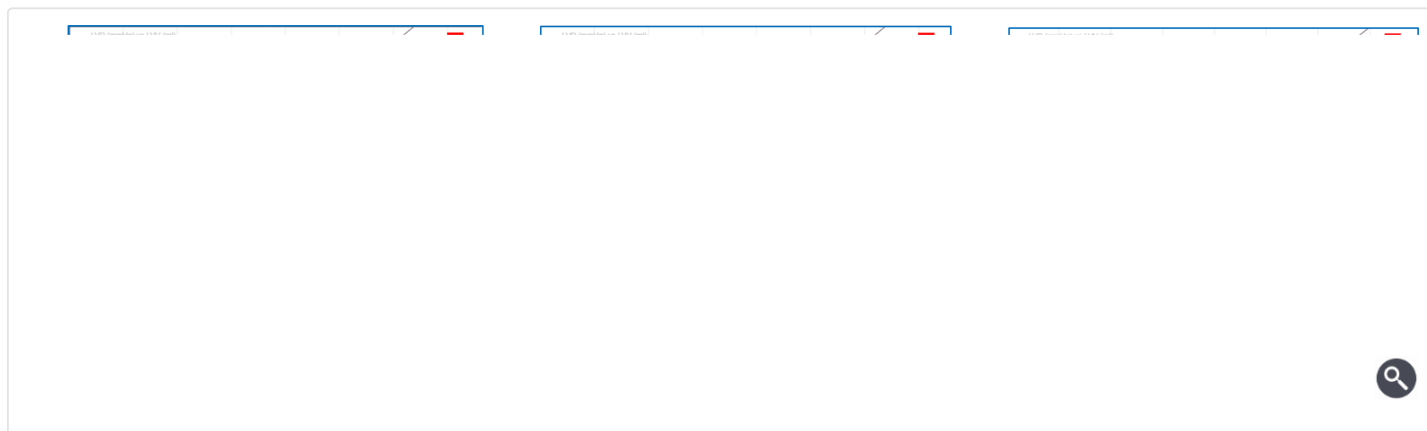
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Practice Exam - 1

Question 83 of 117



A 45-year-old man with known non-ischemic dilated cardiomyopathy LVEF of 20%, moderate mitral valve insufficiency, secondary severe pulmonary hypertension, and chronic kidney disease, presents with acute viral respiratory infection. He required mechanical ventilation. Due to refractory hypoxia and complicated cardiogenic shock with hypotension on 3 pressors, you are deciding on a temporary mechanical circulatory support device.



Which set of three devices produces the hemodynamic changes in the pressure volume loops (gray loops) illustrated in the figure from left to right?

- ☒ **A.** IABP, Impella, ECMO
- ☐ **B.** Impella, ECMO, IABP
- ☐ **C.** IABP, ECMO, Impella
- ☐ **D.** ECMO, Impella, IABP

Testing Point: The AHFTC must understand the hemodynamic changes produced by different mechanical circulatory support devices.

Answer: A. IABP, Impella, ECMO

IABP – positioned in descending aorta deflates in systole and inflates in diastole. There is augmentation of BP in diastole and peak coronary flow increases. During systole, the balloon pump reduces afterload and systolic BP and increases SV, but it only modestly enhances C.O. Balloon pump is more about increasing MAP and coronary flow than unloading LV.

Impella – percutaneous LVAD – for short-term use. Impella pumps blood continuously out of LV into the ascending aorta, increasing forward flow. It is a catheter mounted trans-valvular non-pulsatile axial blood flow device. As one increases flow on Impella, PV loop shifts progressively leftward, indicating greater LV unloading and greater uncoupling between increasing aortic pressure and decreasing peak LV pressure. PV loop area correlates with myocardial work and decreases with Impella support. This direct unloading also results in decreased LAP, wedge pressure and PAP. Impella also decreases native heart work and wall stress and improves systemic perfusion.

V-A ECMO – provides full cardio- and pulmonary support for the patients whose heart and lungs can not provide adequate physiological support. ECMO withdraws deoxygenated blood from RA / venous system and returns oxygenated blood to the arterial system using a centrifugal, non-pulsatile pump. ECMO utilizes a blood gas exchange unit, a membrane oxygenator, to help to normalize acid-base abnormalities. As one increases ECMO blood flow, native SV decreases and PV loop gets narrower; systemic BP increases and the work of the heart is done by ECMO device.

The primary hemodynamic effect of ECMO is increased afterload and blood pressure but the device does not unload LV. LV has to overcome increasing afterload caused by ECMO to eject blood. Since LVEDP and wedge pressure increase with ECMO, sometimes LV may begin to dilate and become more dysfunctional, and thus need to be unloaded. The shape of the PV loop is narrower and the loop moves up and later to the right along the EDPVR curve. ECMO increases central aortic pressure and improves coronary perfusion. ECMO is not designed to unload the LV unlike Impella. The different mechanical circulatory support devices produce specific changes in the PV loops